n = power

1. Differentiation :- at Instantaneous

• Change in something with respect to something.

Differentiation' $\frac{dx}{dt}$ = change in "x" with due to " t "

• 6 Laws •

1) $\frac{d(x)^n}{dx} = n(x^{n-1})$

Ex :- $\frac{dx^5}{dx} = 5x^4$

Ex :- $\frac{dy^3}{dy} = 3y^2$

Ex :- $\frac{dy^{-3}}{dy} = -3y^{-3-1}$

$= -3y^{-4}$

$\frac{dm^{\frac{3}{2}}}{dm} = \frac{3}{2} m^{\frac{3}{2}-1}$

$= \frac{1}{2}$

$= \frac{3}{2} m^{\frac{1}{2}}$

n: power

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$$\left[1.)\quad \frac{d(x)^n}{dx} = n(x^{n-1}) \right]$$

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2. $\boxed{\dfrac{dx}{dx} = 1}$

3. $\dfrac{d(\text{constant})}{dt} = 0$

Ex:-1) $\dfrac{d(3)}{dt} = 0$

Ex:-2) $\dfrac{d(\pi)}{dt} = 0$

Ex:-3) $\dfrac{d(\sin 30^\circ)}{dt} = 0.$

$$\frac{dx^n}{dx} = n(x^{n-1})$$

4 $\frac{d(\text{constant} \times x^n)}{dx} \Rightarrow \text{constant}\left(\frac{dx^n}{dx}\right)$

Ex: $\frac{d(3x^3)}{dx} \Rightarrow 3\left(\frac{dx^3}{dx}\right)$

$= 3 \times 3x^2$

$= 9x^2$

Ex: $\frac{d(4y^5)}{dy} = 4 \times \left(\frac{dy^5}{dy}\right)$

$= 4 \times 5y^4$

$= 20y^4$

Ex: $\frac{d\,5x^{-2}}{dx} \Rightarrow 5\left(\frac{dx^{-2}}{dx}\right)$

$= 5 \times -2x^{-2-1}$

$= 5 \times -2x^{-3}$

$= -10x^{-3}$

4. $\frac{d(m \pm n)}{dx} = \frac{dm}{dx} + \frac{dn}{dx}$

Ex:- $\frac{d(x^3+x)}{dx} \Rightarrow \frac{dx^3}{dx} + \frac{dx}{dx}$

$\Rightarrow 3x^2 + 1$

$= 3x^2 + 1$

Ex:- $\frac{d(y^2+y-y^{-2})}{dy} = \frac{dy^2}{dy} + \frac{dy}{dy} - \frac{dy^{-2}}{dy}$

$= 2y^{1} + 1 - (-2y^{-3})$

5. Product Rule:-

$$\frac{d(m \times n)}{dx} = \frac{dm}{dx} \times n + \frac{dn}{dx} \times m$$

6. Some Variables:-

$$\frac{d(\sin x)}{dx} = \cos x$$

$$\frac{d(\cos x)}{dx} = -\sin x$$

Ex:- $\frac{d(x^2 \sin x)}{dx} = \frac{dx^2}{dx} \times \sin x + \frac{d(\sin x)}{dx} \times x^2$

$\Rightarrow 2x \sin x + \cos x \times x^2$

Ex:- $\frac{d(y^4 \cos y)}{dy} \Rightarrow \frac{dy^4}{dy} \times \cos y + \frac{d(\cos y)}{dy} \times y^4$

$= 4y^3 \times \cos y + -\sin y \times y^4$

6. Dividend Rule:-

$$\frac{d(m/n)}{dx} = \frac{\frac{dm}{dx} \times n - \frac{dn}{dx} \cdot m}{n^2}$$

Ex. $\frac{d(x^2/\sin x)}{dx} \Rightarrow$

$$\frac{\frac{dx^2}{dx} \times \sin x - \frac{d\sin x}{dx} \times x^2}{\sin x^2}$$

$$\Rightarrow \frac{2x \cdot \sin x - \cos x \cdot x^2}{(\sin x)^2}$$

③ $$\frac{d\left(\frac{\cos x}{x^5}\right)}{dx}$$

$$= \frac{d(\cos x)}{dx} \times x^5 - \frac{d x^5}{dx} \times \cos x .$$

$$\Rightarrow \frac{-\sin x \cdot x^5 - 5x^4 \cdot \cos x}{\cancel{\cos^2 x}\ (x^5)^2} \Rightarrow x^{10}$$

$\times x^2$